Dylan Thornburg

SANTA CLARA UNIVERSITY	ELEN 153	Shoba Krishnan
Laboratory #3: CMOS Inverter		

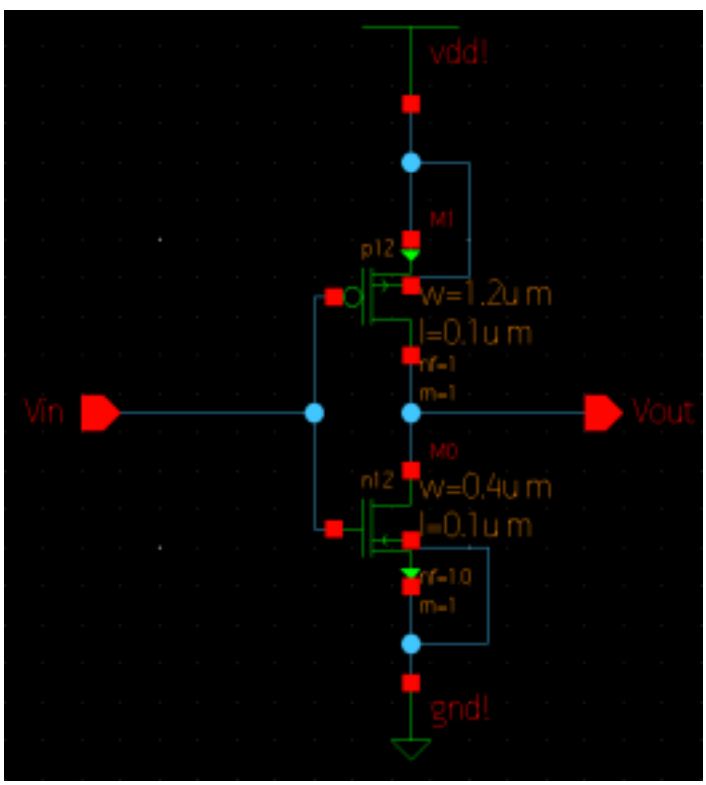
I. OBJECTIVES

- To learn to create and edit symbols and testbenches in Synopsys Custom Compiler
- To capture a CMOS inverter schematic and symbol
- To simulate the Voltage transfer characteristic (VTC) of the inverter
- To study the delay and power of a CMOS inverter

II. LAB PROCEDURE

1. Schematic and Symbol

Follow Section V of the Synopsys Tutorial pages 20-24 to create the schematic and symbol[1] of the CMOS inverter.

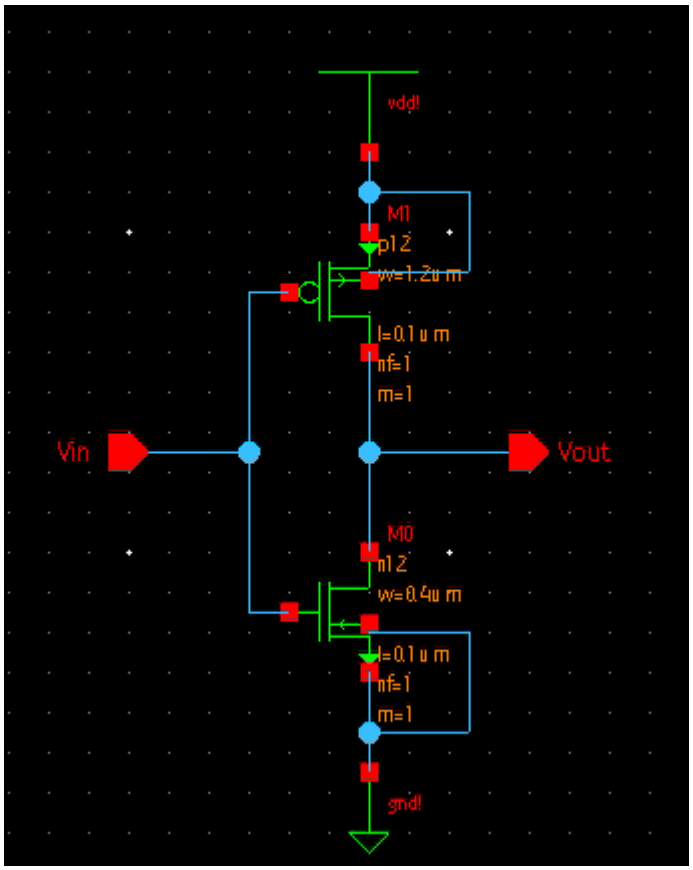


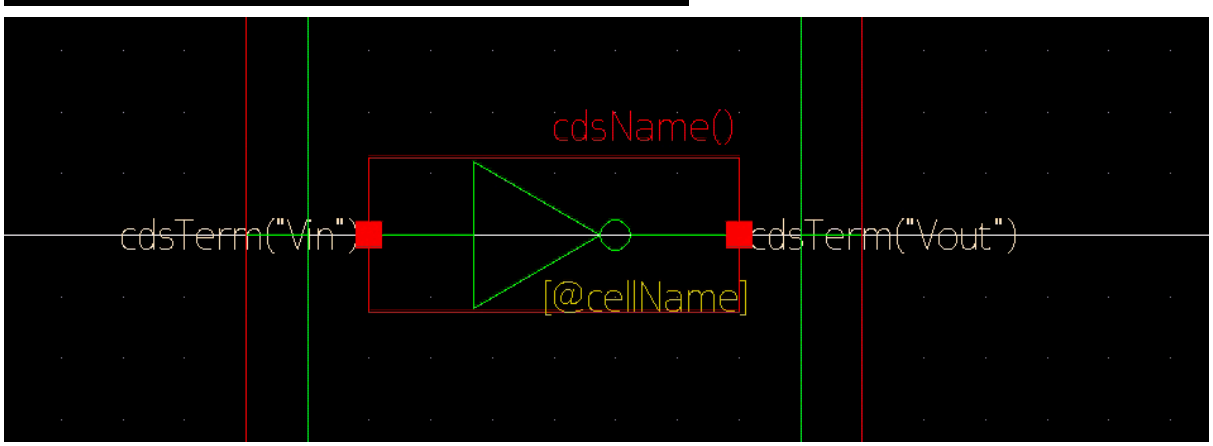
[1] Symbols are useful for hierarchical description of large circuits, analogous to subroutines within a larger program. It enables 're-use' of a basic building-block within a larger circuit. In the corresponding netlist description, symbols become subcircuits.

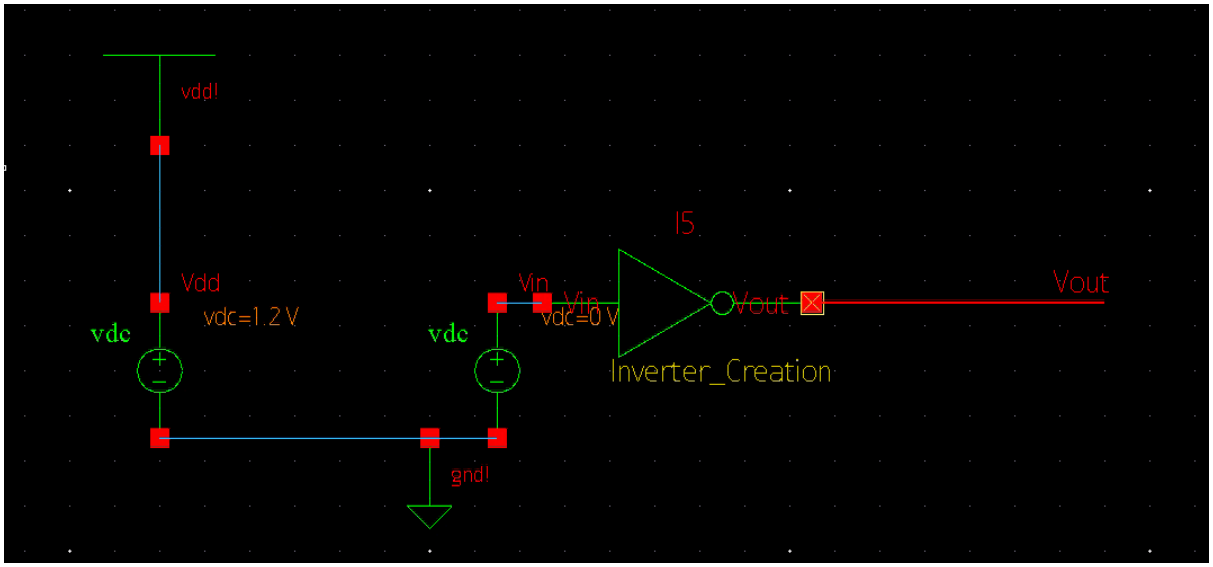
2. Simulations

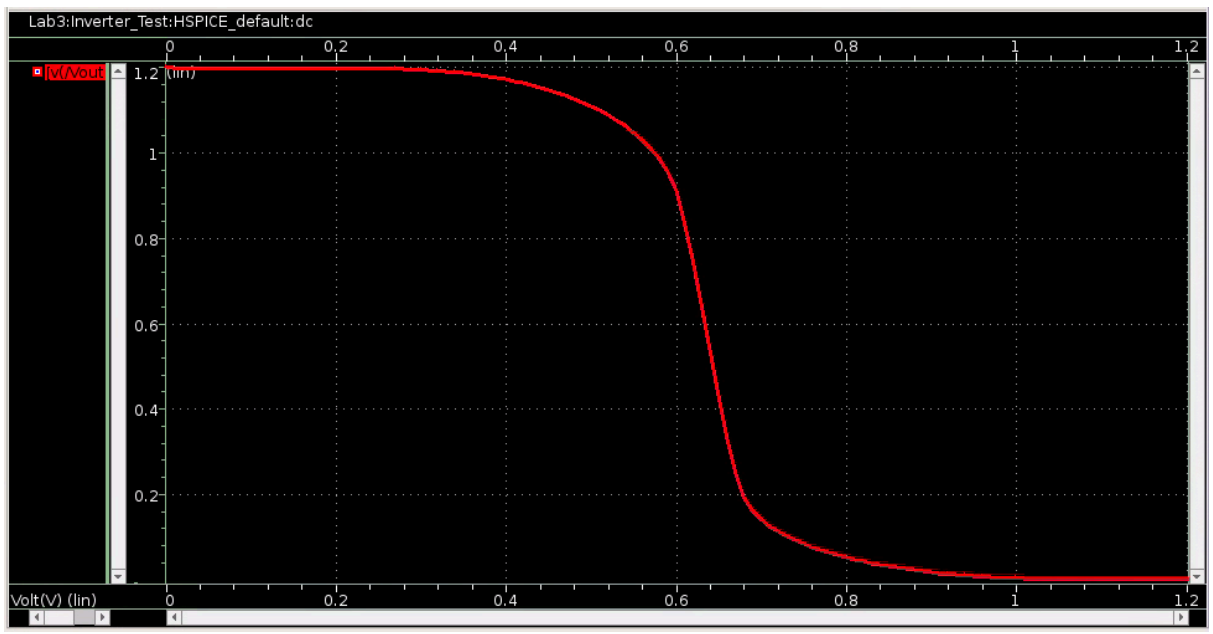
Continue Section V of the Synopsys Tutorial pages 24-32 to perform VTC simulations as well as transient simulations connected to a 1pF capacitor in testbench.

RESPONSE:





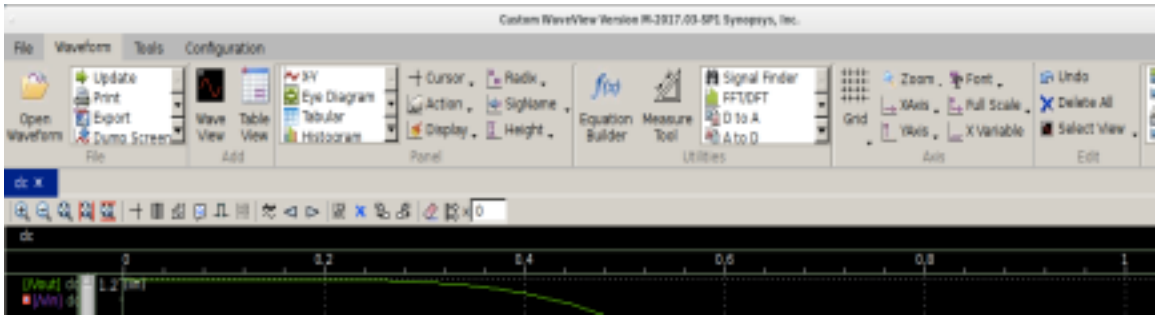


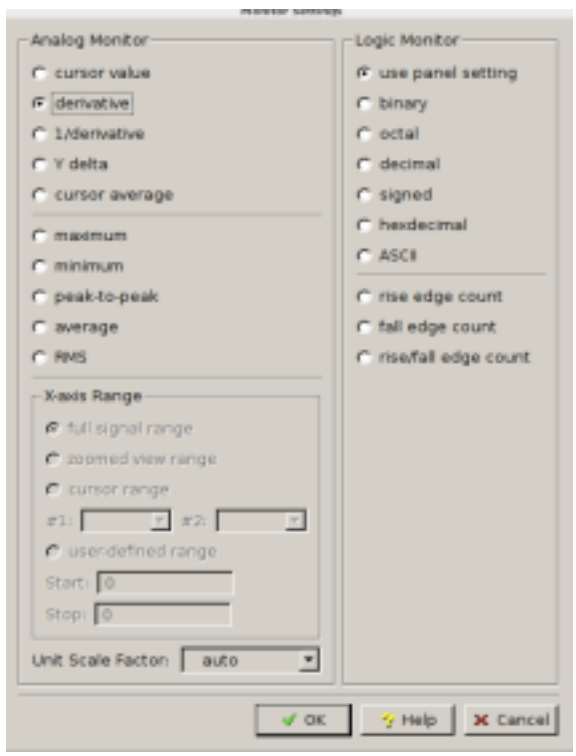


Part 1: Inverter VTC

From the VTC (**DC Simulation**) result find the following:

- Inverter threshold voltage V_M graphically, where VM = V(IN) = V(OUT).
- V_{IH} ,V_{IL} , V_{OL} and V_{OH} of the inverter by finding the value of the input voltage, at which the slope of Vout = -1. To do that, in Custom WaveView, follow the steps below:
    1) Add a cursor to your waveforms, by selecting the icon below.
    2) Select the Waveform Monitor tool and select derivative.
    3) Find the Vin value for when Vout slope is -1

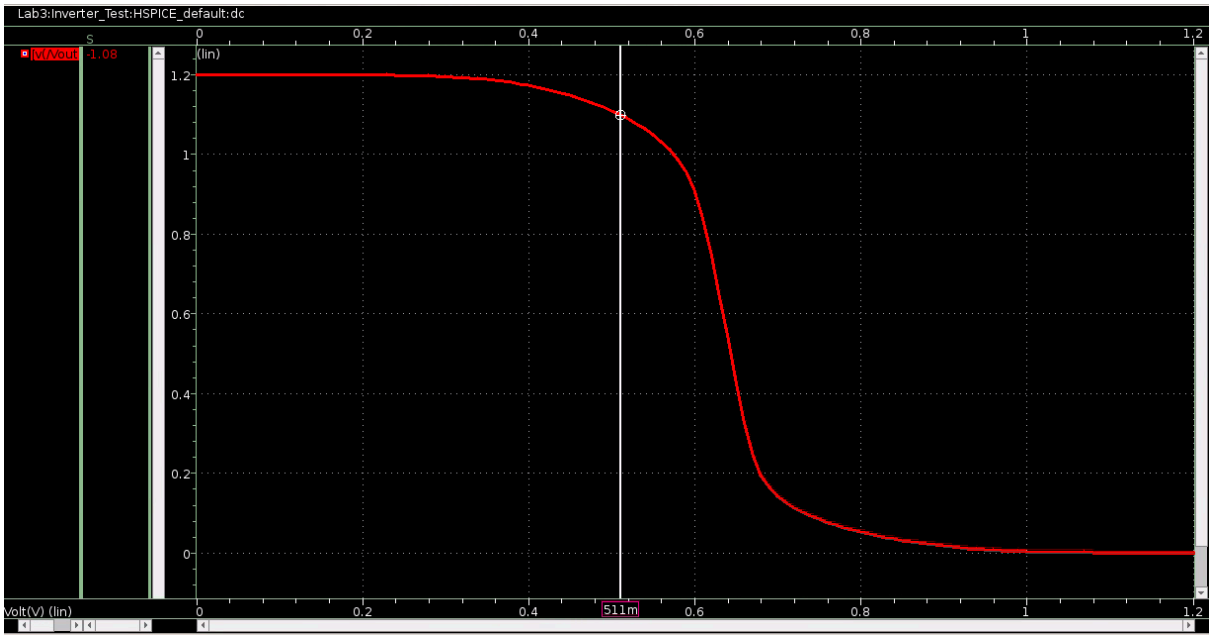




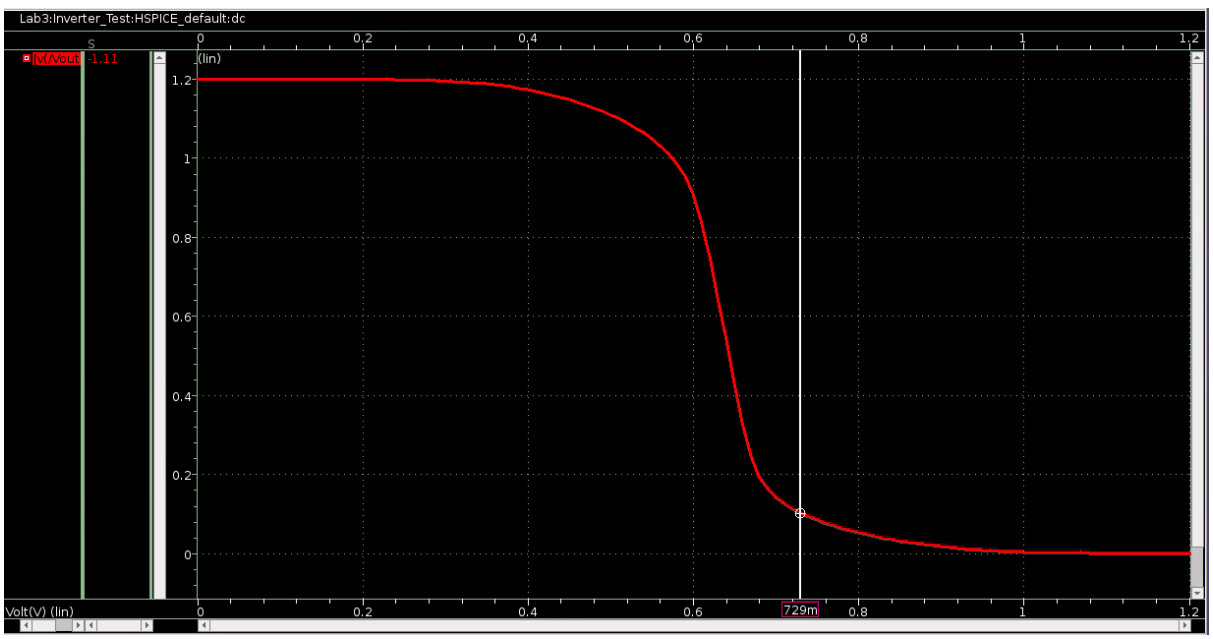
[1]

Symbols are useful for hierarchical description of large circuits, analogous to subroutines within a larger program. It enables 're-use' of a basic building-block within a larger circuit. In the corresponding netlist description, symbols become subcircuits.

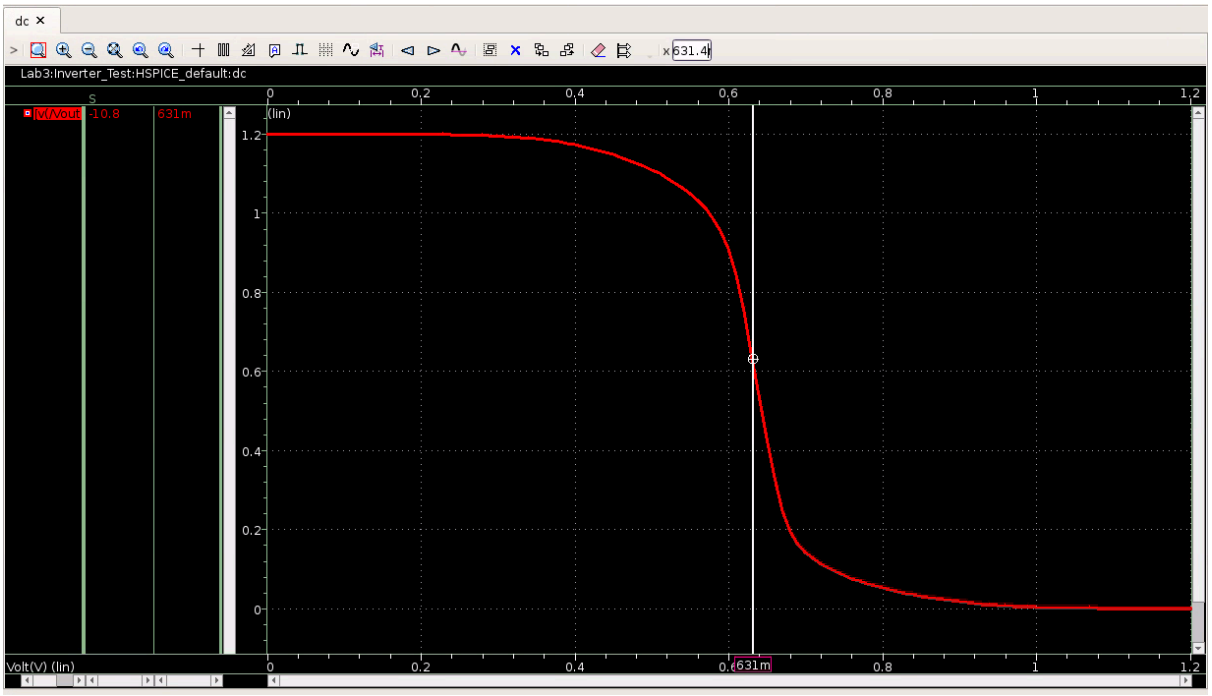
- Note the values of V_M, V_{IH} ,V_{IL} , V_{OL} and V_{OH} along with the proof of measured screenshots.



The Vih is around .511V. The Voh is ~1.2V.



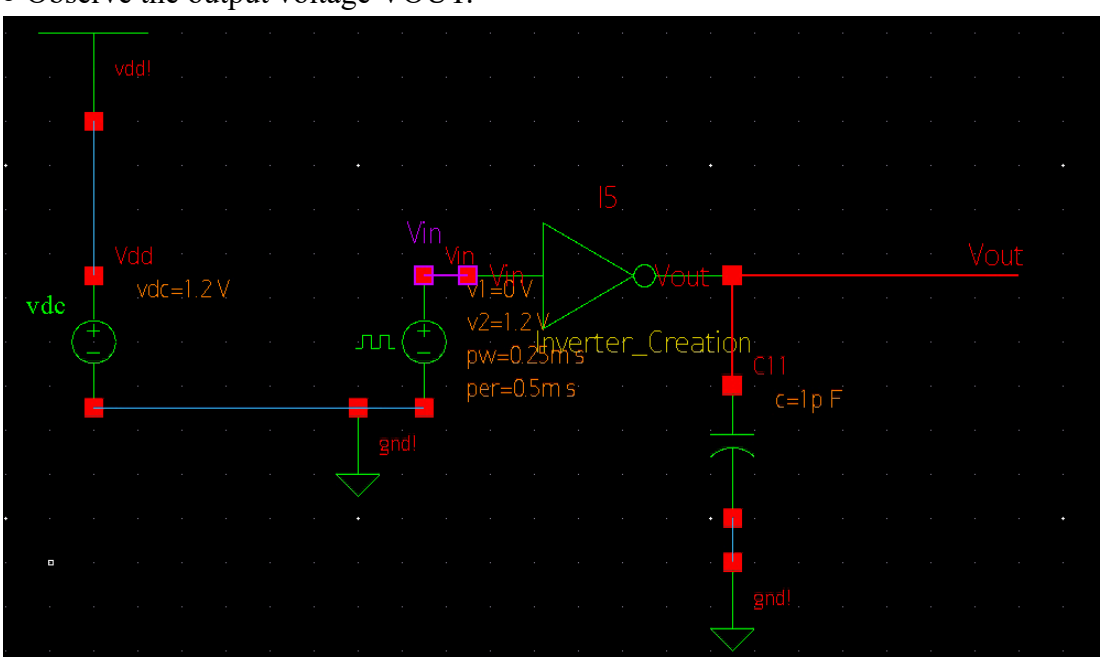
The Vil is around .729V. The Vol is ~0V

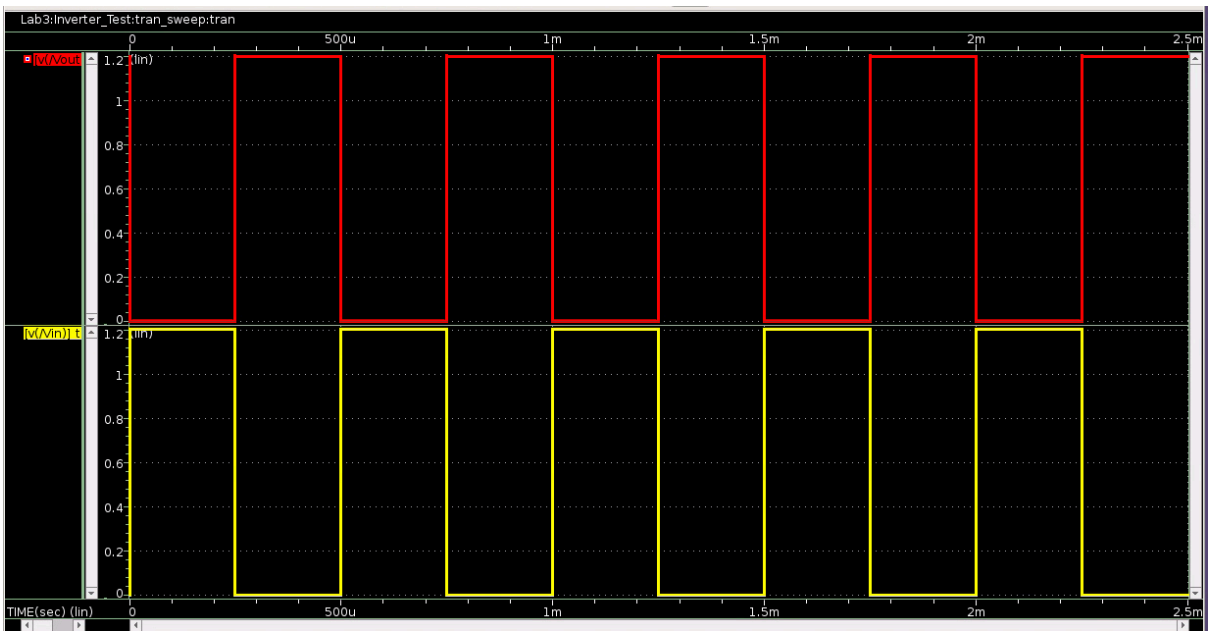


The Vm is around .631V

Part 2: Noise margins

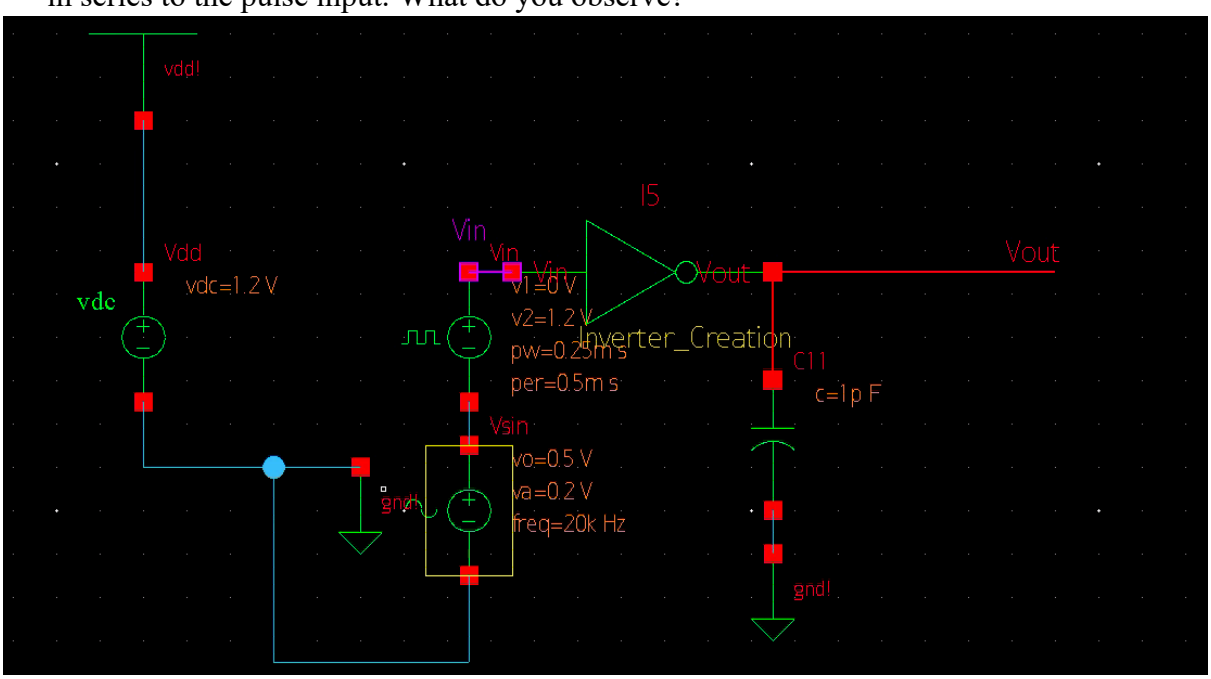
- Run a transient simulation with a pulse input for VIN from 0 to VDD, pulse width of 250uS, period 500uS with rise/fall time of 10nS.
- Observe the output voltage VOUT.

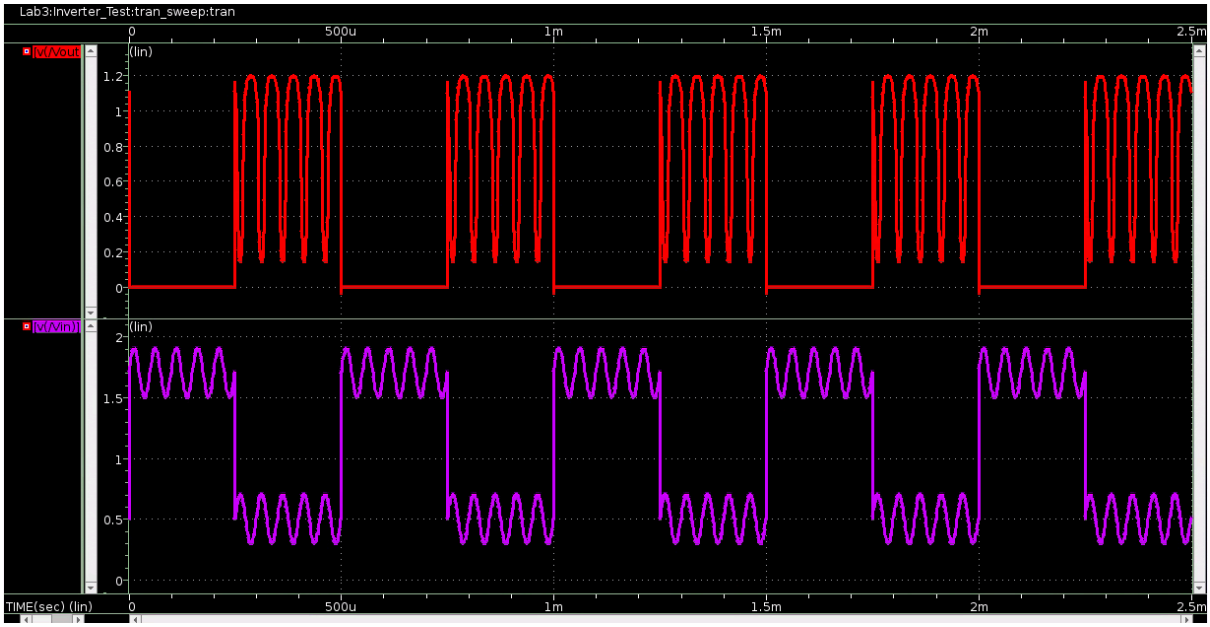




Expected Vout behavior. It inverts.

- Add a sinusoidal voltage of offset 500mV, amplitude 200mV, Frequency 20K Hz in series to the pulse input. What do you observe?





What a weird wave. The input is a combination of the square and sin wave, and the output is the inverse of it (sort've).The output is also limited to 1.2V max.

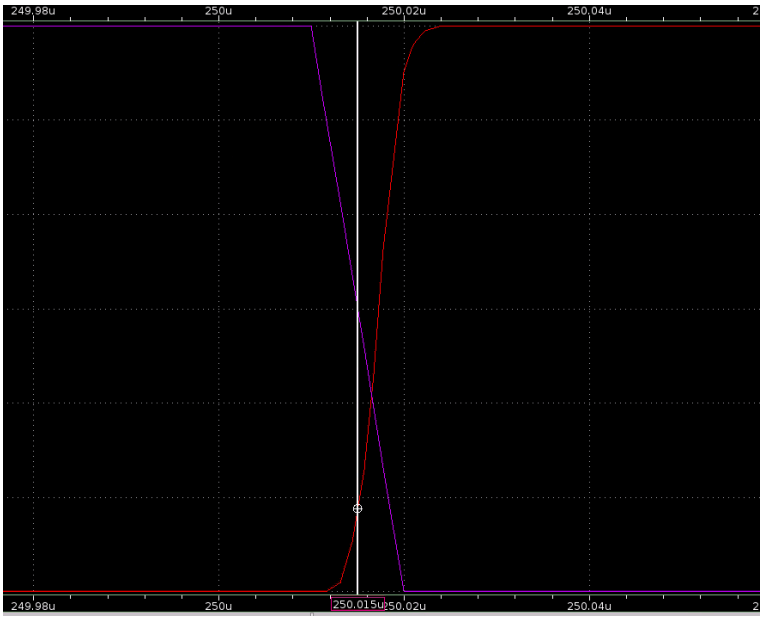
Part 3: Propagation delay

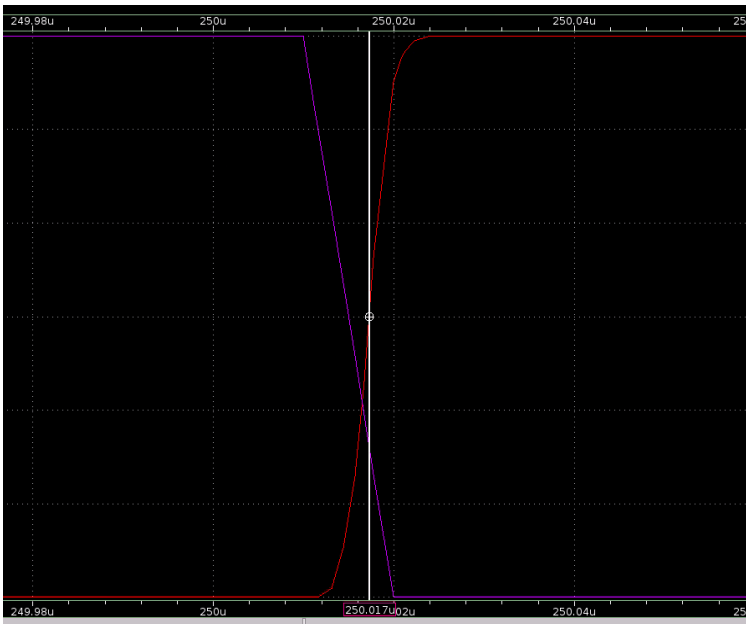
- Run a transient simulation with a pulse input for VIN from 0 to VDD, pulse width of 250uS, period 500uS with rise/fall time of 10nS.
- Observe the output voltage VOUT. Measure the propagation delay t_{pLH} and t_{pHL} by measuring from 50 of the input to 50% of the output.

Low to High:

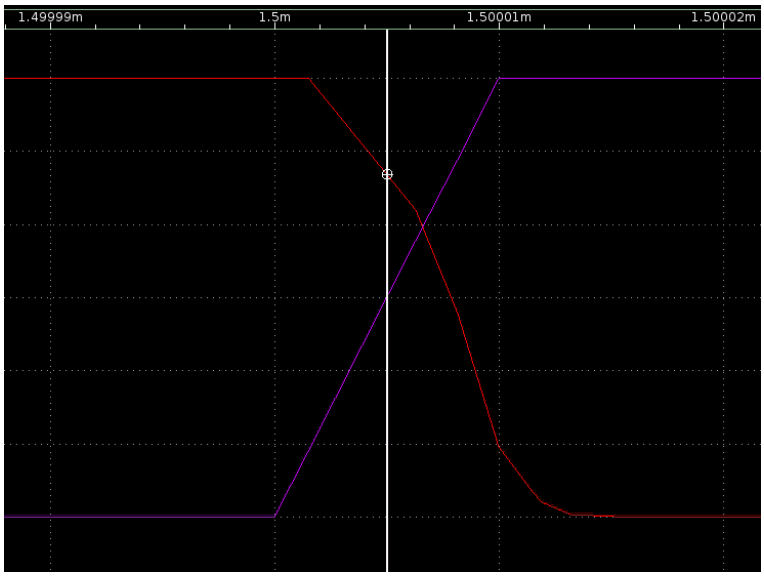
50% Vin at 250.015u, 50% Vout at 250.017311u

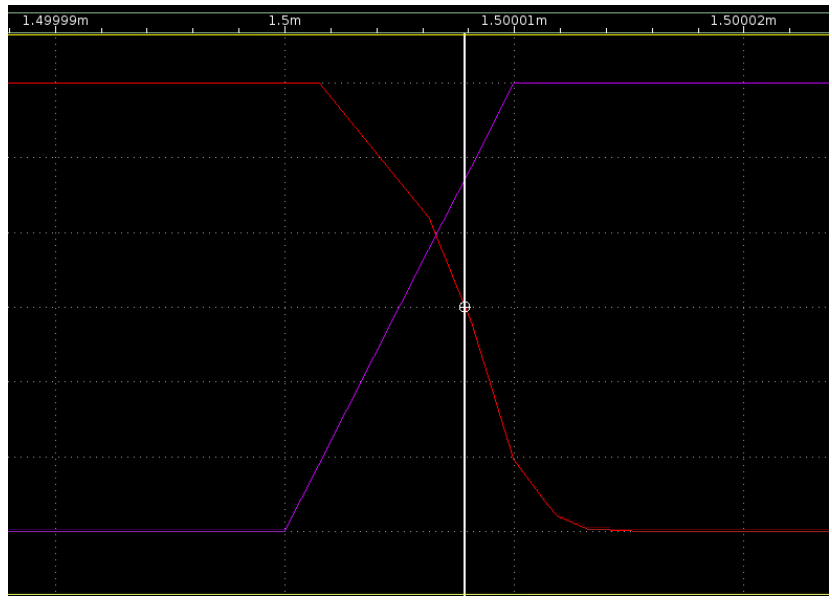
250.017311u-250.015u= 2.311n. Prop delay tpLH is 2.311ns





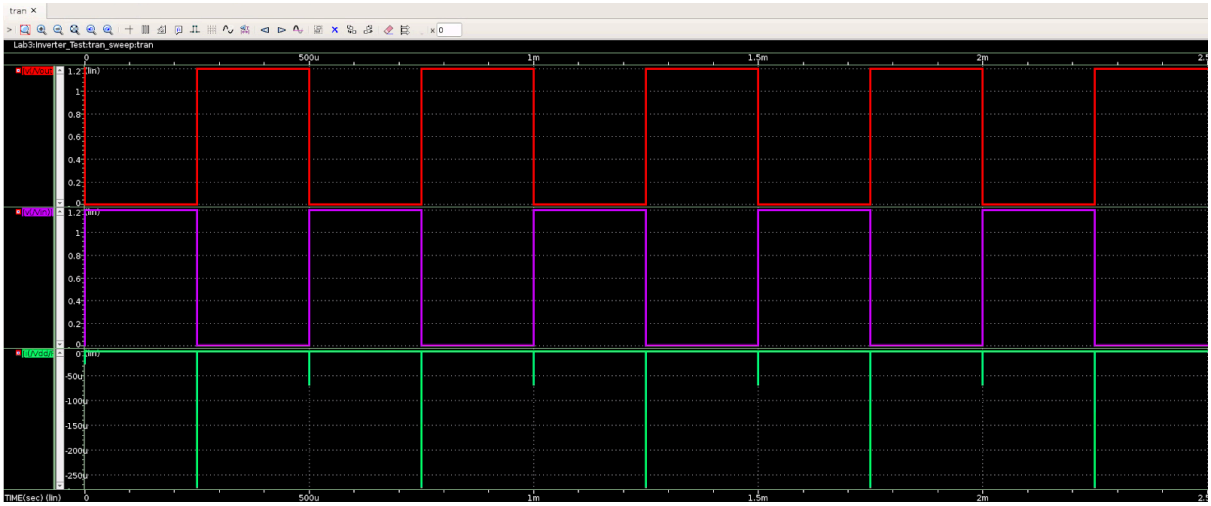
High to Low:
50% Vin at 1500.005u, 50% Vout at 1500.00786u
1500.00786-1500.005= 2.86n. Prop delay tpHL is 2.86ns





● Observe the current drawn from the PMOS and NMOS transistors. What device works on pulling the output to VDD and what works on pulling it down to ground?
The PMOS pulls the output to VDD while the NMOS pulls the output to ground.

● Plot the current from the power supply VDD. What do you notice about the current from VDD. Explain what the various sections are and why this happens.
The currents are very sudden and represent the short times when the NMOS and PMOS are on. Also we get a higher current flow when the cap is charging.



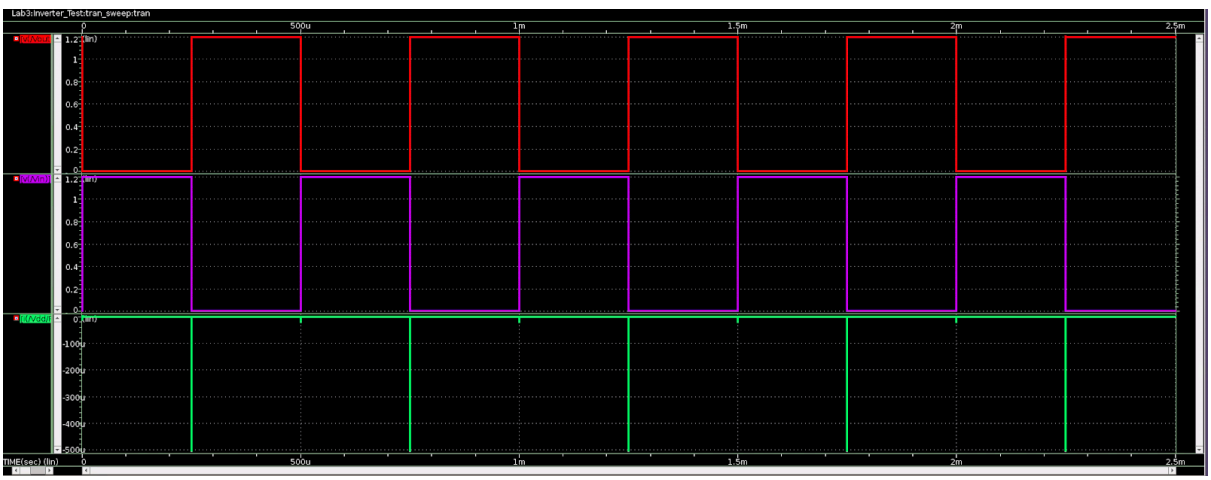
● Change the rise/fall time to 1ns.

● Observe the current drawn from the PMOS and NMOS transistors. What device works on pulling the output to VDD and what works on pulling it down to ground?

The PMOS pulls the output to VDD while the NMOS pulls the output to ground.

● Plot the current from the power supply VDD. What do you notice about the current from VDD. Explain what the various sections are and why this happens.

We see the same sections again, though the abs value of the current through the PMOS (charging) greater than before, and the abs value of current through the NMOS (discharging) is less than before.

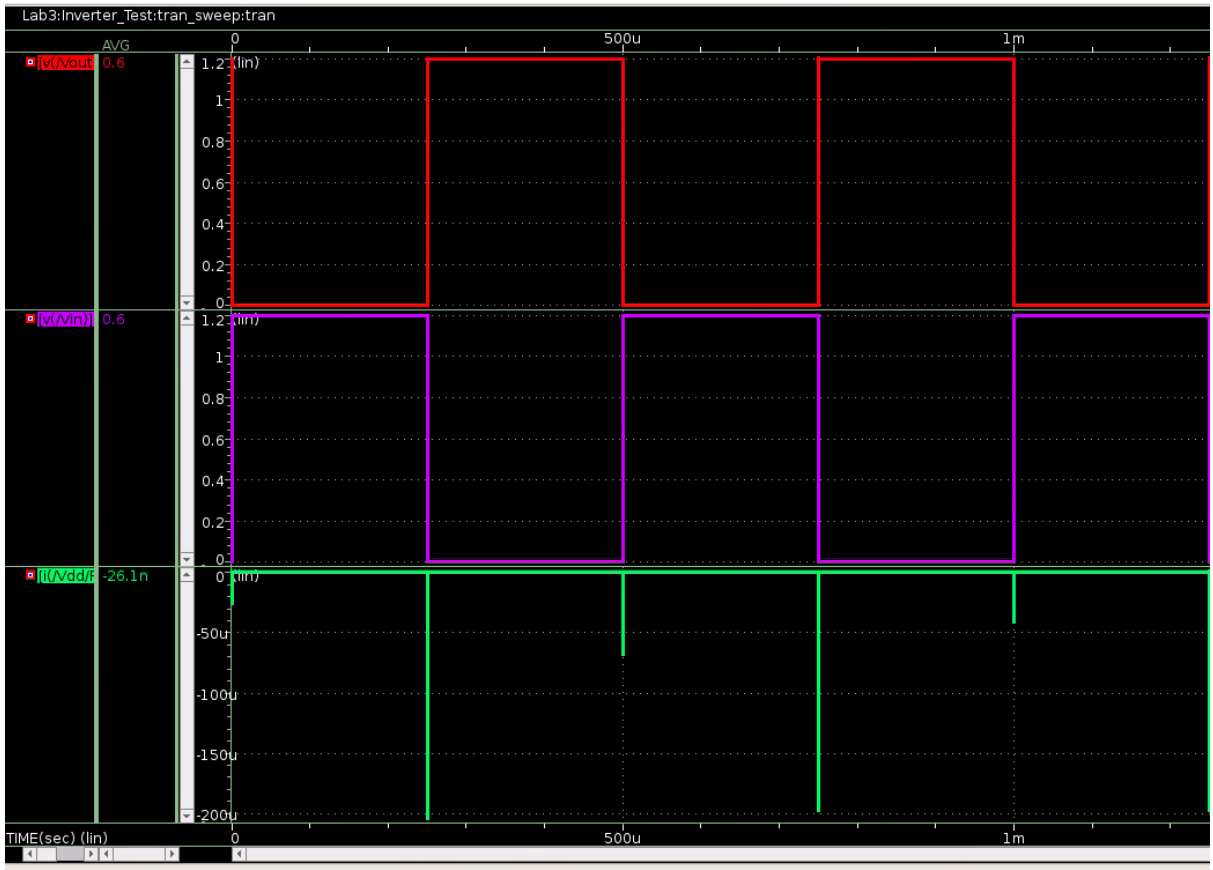


Part 4: Power

● Run a transient simulation with a pulse input for VIN from 0 to VDD, pulse width of 250uS, period 500uS with rise/fall time of 10nS.

● Measure the current drawn from VDD supply. What is the average value of this current.

The avg is -26.1 nA which is effectively 0 (but still not quite!)



III. REPORT

Write a short laboratory report that details all the work done. Describe the objective and procedures of this lab with your own words. The lab report should contain the following:

a) All schematics used in your lab.
b) Screenshots of all Simulation setups in SAE
c) All Simulation Results, including waveforms and any required values/measurements.
d) Answer any questions in the lab assignment
e) Conclusions

Conclusions:
A CMOS inverter is a good way to invert a signal. The delay is pretty small and the circuit is overall quite efficient. If I wanted to make the time delays match, I would increase the size of the PMOS a bit.